MANIPAL INSTITUTE OF TECHNOLOGY
BENGALURU
(A constituent unit of MAHE, Manipal)

School of Computer Engineering
Machine Learning Lab (CSE_3146)
Mini Project Proposal

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Dept/Branch: 3rd year AI-B

Project Title: Emotion Reader

1. Background / Motivation:

The ability to understand and interpret human emotion is a cornerstone of effective communication, yet it's often subtle and difficult to quantify. From improving mental wellness and customer service to tailoring educational content, there is a significant gap in being able to accurately and objectively read and respond to emotional cues. The motivation for this project is to bridge this gap by creating an accessible tool that uses technology to provide real-time insights into human emotional states.

2. Objectives:

Our primary objectives are to:

1. Develop an application capable of **Emotion Recognition**, which can detect and classify the emotional state of a user from a live video feed.
2. Leverage cutting-edge **Machine Learning** techniques to build a robust model that can analyze and interpret subtle facial cues with a high degree of accuracy.
3. Create a clean, intuitive, and responsive user interface for real-time visualization of the detected emotions, ensuring a smooth and useful user experience.

3. Proposed Methodology / Approach:

The app will be a single-page web application using a pre-trained convolutional neural network (CNN) model for Facial Expression Analysis. The model, trained on a publicly available dataset, will process webcam video in Real-time Processing to predict emotions and a confidence score. This data will be shown in a simple user interface.

4. **Expected Outcomes / Scope:**

The project is expected to deliver a functional web-based prototype capable of the following:

- Capturing video input from a webcam.
- Detecting and classifying a user's emotion with a high level of accuracy.
- Providing a Real-time Processing display of the detected emotion and its confidence level.
- Presenting a simple, yet elegant, user interface that is both functional and easy to navigate. This project will serve as a foundational step toward more advanced applications in emotional intelligence technology

Keywords: Emotion Recognition , Facial Expression Analysis, Convolutional Neural Network (CNN), Real-time Processing, Computer Vision